Discrete Mathematics Midterm Exam Collection

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1 2022 Midterm Exam

Discrete Mathematics

Midterm Exam, 2022

1. Express the statement $\lim_{x \to x_0} f(x) = A$ using quantifiers.

2. Prove or disprove: a standard international chessboard with its four corner squares removed can be tiled by dominoes.

3. Prove or disprove that $p \to (q \to r)$ and $q \to (p \wedge r)$ are logically equivalent.

4. For nonempty sets A and B, prove that $A \times B = B \times A$ if and only if $A = B$.

5. Let f be a function from A to B, and let $S, T \subseteq A$. Prove that

(1) $f(S \cup T) = f(S) \cup f(T)$.

(2) $f(S \cap T) \subseteq f(S) \cap f(T)$.

6. Let A be a countable set. If there is an onto function $f : A \to B$, prove that B is countable.

7. For the polynomial algorithm flowchart in the original paper, determine the number of addition and multiplication operations.

8. Prove that

$$1^k + 2^k + \cdots + n^k \text{ is } O(n^{k+1}).$$

9. Find $123^{1001} \pmod{101}$**.**

10. Solve the system of congruences

$$x \equiv 5 \pmod 6, \qquad x \equiv 3 \pmod{10}, \qquad x \equiv 8 \pmod{15}.$$

11. Use strong induction to prove that postage of every amount at least 18 cents can be formed using only 4-cent and 7-cent stamps.

Bonus Question 12.

(a) Give a recursive definition of binary trees.

(b) Give an algorithm, written in pseudocode, that visits every node of a binary tree.

(c) Prove the correctness of your algorithm.

2 2024 Midterm Exam (WYL)

Discrete Mathematics, Spring & Summer 2024

Midterm Exam (100 pts)

1. (33 points) Determine whether the following statements are true or false. If it is true, write a ✓; otherwise write a × in the blank before the statement.

( ) (1) The proposition "$-1 + 1 = 3$ if and only if $2 + 2 = 3$" is true.

( ) (2) The proposition $((p \to q) \wedge \neg p) \to \neg q$ is a tautology.

( ) (3) The following system specifications are consistent.
Whenever the system software is being upgraded, users cannot access the file system.
If users can access the file system, then they can save new files.
If users cannot save new files, then the system software is not being upgraded.

( ) (4) If $P(A) \in P(B)$, then $A \in B$ ($P(S)$ is the power set of S).

( ) (5) Suppose $A = \{a, b, c\}$. Then $\{\{a\}\} \subset P(A)$.

( ) (6) $A \oplus A = A$ ($\oplus$ is the symmetric difference of two sets).

( ) (7) Let f be a function from A to B, and let $S, T \subseteq A$. Then

$$f(S \cup T) \subseteq f(S) \cup f(T).$$

( ) (8) Let $\mathbb{R}$ be the set of real numbers. Then $|\mathbb{R}| = |\mathbb{R} \times \mathbb{R}|$.

( ) (9) The set of all finite subsets of the set of positive integers is uncountable.

( ) (10) $\sum_{j=1}^{n}(j^3 + j)$ is $O(n^3)$.

( ) (11) In a group of five people, where any two people are either friends or enemies, there are either three mutual friends or three mutual enemies.

2. (27 points) Fill in the blanks.

(1) (2 pts) The full disjunctive normal form of $(p \wedge q) \vee (r \wedge q) \vee \neg p$ is

_____.

(2) (2 pts) The power set of $\{\emptyset, \{\emptyset\}\}$ is _____.

(3) (2 pts) Suppose $g : A \to B$ and $f : B \to C$, where $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, $C = \{2, 6, 10\}$, and

$$g = \{(1, b), (2, a), (3, a), (4, b)\}, \qquad f = \{(a, 10), (b, 7), (c, 2)\}.$$

Then $f \circ g =$ _____.

(4) (2 pts) How many different truth tables of compound propositions are there that involve the propositional variables p, q, s? _____

(5) (2 pts) Arrange 2.5^n, n^{100}, $(\log n)^3$, $n \log n$, $(n!)^2$, and $n^{99} + n^{98}$ so that each function is big-O of the next. _____

(6) (2 pts) Suppose $|A| = 20$ and $|B| = 24$. The number of one-to-one functions $f : A \to B$ is _____. If $|A| = 24$ and $|B| = 20$, the number of one-to-one functions $f : A \to B$ is _____.

(7) (3 pts) You have 10 books and 4 labeled boxes. In how many ways can you put the books into the boxes if
(a) the books are distinct? _____
(b) the books are identical? _____
(c) the books are identical and no box can be left empty? _____

(8) (2 pts) The number of ways to put 7 of the 8 letters in `CHEMISTS` in a row is

_____.

(9) (2 pts) The next larger permutation in lexicographic order after 3254761 is

_____.

(10) (2 pts) The next larger 4-combination of $\{1, 2, 3, 4, 5, 6, 7, 8\}$ after $\{1, 2, 3, 5\}$ is

_____.

(11) (2 pts) The number of bit strings of length n, where $n \geq 6$, containing exactly three occurrences of 01 is _____.

(12) (2 pts) Let $Y(n)$ be the number of strings of n letters of the English alphabet with no adjacent vowels. Give a recurrence for $Y(n)$ and enough initial conditions.

_____

(13) (2 pts) A particular solution of $a_n = 2a_{n-1} + n - 2^n$ has the form

_____.

3. (5 points) Determine whether there exists an onto function from S to $P(S)$, where S is a set and $P(S)$ is its power set. Justify your answer.

4. (5 points) Consider the following argument:
She is a Math Major or a Computer Science Major.
If she does not know discrete math, she is not a Math Major.
If she knows discrete math, she is smart.

She is not a Computer Science Major.
Therefore, she is smart.
Let p: "She is a Math Major," q: "She is a Computer Science Major," r: "She knows discrete math," and s: "She is smart."
1) Write the formal form of the argument.
2) Determine whether the argument is valid and justify your answer.

5. (10 points) Show that in any set of $n+1$ positive integers not exceeding $2n$, there must be two that are relatively prime.

6. (10 points) Prove the identity using a combinatorial argument:

$$\sum_{k=0}^{r} \binom{n+k}{k} = \binom{n+r+1}{r}, \qquad n, r \in \mathbb{Z}^+.$$

7. (10 points) Suppose a vending machine accepts \$1 coins, \$1 bills, \$2 bills, \$5 bills, and \$5 coins. Write a recurrence relation for the number of ways to put n dollars into the machine, assuming that order matters. Include the initial conditions.

3 2024 Midterm Exam (LLF)

Discrete Mathematics, Spring & Summer 2024

Midterm Exam (100 pts)

1. (10 points) Determine whether the following statements are true or false.

( ) (1) $2^{2022} \equiv 1 \pmod 7$.

( ) (2) Let m be a positive integer and let a, b, c be integers. If $ac \equiv bc \pmod m$, then $a \equiv b \pmod m$.

( ) (3) The set $\mathbb{Z}$, with the usual $\leq$ ordering, is well-ordered.

( ) (4) In any group of six people, where each pair consists of either two friends or two enemies, there are either three mutual friends or three mutual enemies.

( ) (5) 7 is an inverse of 5 modulo 17.

2. (30 points) Fill in the blanks.

(1) What form does a particular solution of

$$a_n = 4a_{n-1} - 4a_{n-2} + F(n)$$

have when $F(n) = (n^2 + 1)2^n$? _____

(2) Find the number of permutations of the letters in `multiplication`.

_____

(3) Suppose $|A| = 6$ and $|B| = 5$. Find the number of one-to-one functions $f : A \to B$.

_____

(4) A computer randomly prints three-digit codes with no repeated digits, allowing leading zeroes. What is the minimum number of codes that must be printed to guarantee that at least six of the codes are identical? _____

(5) How many solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 21$, where each $x_i \geq 2$ is an integer? _____

(6) Find the coefficient of x^{12} in the power series of $\dfrac{x^3}{(1+4x)^2}$. _____

(7) An office manager has four employees and nine reports to be done. In how many ways can the reports be assigned to the employees so that each employee has at least one report to do? _____

(8) Use Fermat's little theorem to find $9^{45} \pmod{23}$. _____

(9) Express $\gcd(450, 120)$ as a linear combination of 120 and 450. _____

(10) Suppose $|A| = 4$. Among all binary relations on A, there are _____ antisymmetric relations.

3. (8 points) Let S be the set of positive integers defined recursively by

$$4 \in S, \qquad \text{if } n \in S, \text{ then } 5n + 2 \in S \text{ and } n^2 \in S.$$

(1) Show that if $n \in S$, then $n \equiv 4 \pmod 6$.

(2) Show that there exists an integer $m \equiv 4 \pmod 6$ that does not belong to S.

4. (10 points) Use generating functions to solve

$$a_n = 7a_{n-1} - 10a_{n-2}, \qquad a_0 = 1, \quad a_1 = 1.$$

5. (10 points) There are r red balls, r blue balls, and r white balls. How many ways are there to select 12 balls from the three piles if the number of red balls selected must be even?

6. (12 points) A country uses only \$1 and \$2 coins. Let a_n denote the number of ways to pay n dollars using only these coins, where the order of payment matters and coins of the same value are indistinguishable.

(1) Find a recurrence relation for a_n and give the necessary initial conditions.

(2) Find an explicit formula for a_n by solving the recurrence in part (1).

4 2025 Midterm Exam (WYL)

Discrete Mathematics, Spring & Summer 2025

Midterm Exam (100 pts)

1. (20 points) Determine whether the following statements are true or false. If true, write a ✓; otherwise write a ×.

( ) (1) The proposition "$2 \cdot 3 = 8$ if and only if $3 + 2 = 4$" is true.

( ) (2) The following system specifications are consistent.
Whenever the system software is being upgraded, users cannot access the file system.
If users can access the file system, then they can save new files.
If users cannot save new files, then the system software is not being upgraded.

( ) (3) Suppose $A = \{a, b, c\}$. Then $\{a\} \subset P(A)$.

( ) (4) $\overline{A \cup B} \cup \overline{A} = \overline{A}$.

( ) (5) $A \oplus A = A$, where $\oplus$ is symmetric difference.

( ) (6) The function $f : \mathbb{N} \to \mathbb{N}$ defined by $f(x) = \lceil x \rceil + 1$ is onto.

( ) (7) $(\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}) - (\mathbb{R}^+ \times \mathbb{R}^+ \times \mathbb{R}^+)$ is uncountable.

( ) (8) Let $F = \{f \mid f : \mathbb{N} \to \{0, 1\}\}$. Then F is countably infinite.

( ) (9) $\sum_{j=1}^{n}(j^3 + j)$ is $O(n^4)$.

( ) (10) In a group of five people, where any two people are either friends or enemies, there are either three mutual friends or three mutual enemies.

2. (30 points) Fill in the blanks.

(1) Let $F(A)$ mean "A is finite" and $S(A, B)$ mean "A is a subset of B". The domain consists of all sets. Translate into symbols: Every subset of a finite set is finite.

_____

(2) The prenex disjunctive normal form of

$$(\exists x P(x) \wedge \forall x Q(x)) \to \exists x(\neg P(x) \to Q(x))$$

is _____.

(3) The power set of $\{\emptyset, \{\emptyset\}\}$ is _____.

(4) Suppose $g : A \to B$ and $f : B \to C$, where $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, $C = \{3, 7, 10\}$,

$$g = \{(1, b), (2, a), (3, c), (4, b)\}, \qquad f = \{(a, 10), (b, 7), (c, 3)\}.$$

Then $f \circ g =$ _____.

(5) How many different truth tables of compound propositions are there that involve p and q? _____

(6) Arrange $(1.5)^n$, n^{100}, $(\log n)^3$, $\sqrt{n}\log n$, 10^n, $(n!)^2$, and $n^{99}+n^{98}$ so that each is big-O of the next. _____

(7) Suppose $|A| = 6$ and $|B| = 12$. The number of one-to-one functions $f : A \to B$ is _____. If $|A| = 12$ and $|B| = 6$, the number is _____.

(8) You have 19 cards and 10 labeled envelopes. In how many ways can you put the cards into the envelopes if
(a) the cards are distinct? _____
(b) the cards are identical? _____
(c) the cards are identical and no envelope can be left empty? _____

(9) The number of permutations of the letters in `TATTERED` is _____.

(10) The next larger permutation in lexicographic order after 31528764 is

_____.

(11) The next four larger 4-combinations of $\{1,2,3,4,5,6,7,8\}$ after $\{2,3,4,8\}$ are

_____.

(12) The number of bit strings of length n, where $n \geq 4$, containing exactly two occurrences of 01 is _____.

(13) The number of ordered pairs (A, B) such that A and B are subsets of $S = \{a, b\}$ and $A \subseteq B$ is _____.

3. (10 points) Consider the following argument:
Each student is either male or female.
A student is male if and only if he wears glasses.
Not all students wear glasses.
Therefore, some students are female.
Let the domain consist of all students in a class. Let $F(x)$ mean "x is female," $M(x)$ mean "x is male," and $G(x)$ mean "x wears glasses."
1) Write the formal form of the argument.
2) Determine whether this argument form is valid and justify your answer.

4. (10 points) Suppose f is a function from A to B, where A and B are finite nonempty sets. Show that $|f(S)| = |S|$ for every subset S of A if and only if f is one-to-one.

5. (10 points) Use the well-ordering property to show that every integer greater than 1 has at least one prime divisor.

6. (10 points) Let $s_1, s_2, \ldots, s_{101}$ be 101 bit strings of length at most 9. Prove that there exist two strings s_i and s_j, with $i \neq j$, that contain the same number of 0s and the same number of 1s.

7. (10 points) Prove the identity using a combinatorial proof, where n and r are nonnegative integers with $n \geq r$:

$$P(n+1, r)(n+1-r) = (n+1)P(n, r).$$

Bonus Question (10 points). Show that if $m, n \geq 3$, then

$$R(m, n) \leq R(m, n-1) + R(m-1, n),$$

where $R(m, n)$ is the Ramsey number.

5 2025 Midterm Exam (ZYW)

Discrete Mathematics, Spring & Summer 2025

Midterm Exam (100 pts)

1. (10 points) A game has three propositions p, q, r. Each round you may propose a compound proposition in which each of p, q, r appears exactly once, using only $\neg, \wedge, \vee$ and parentheses. The answer is true or false. The game ends when the truth values of p, q, r are uniquely determined. The respondent is perfectly logical and tries to maximize the number of rounds.
Provide a strategy that ends the game as quickly as possible, and determine the minimum number of rounds required to guarantee that the game ends.

2. (10 points) An ordered pair (a, b) is represented as $\{a, \{a, b\}\}$. Express the following in set notation, without using ordered-pair notation.

(i) $\{1\} \times \{2\}$.

(ii) $\{1, 2\} \times \{2, 3\}$.

(iii) $\emptyset \times \{1, 2\}$.

(iv) $(1, 2) \times \{1, 2\}$.

(v) $\{\emptyset\} \times \{\emptyset, \{\emptyset\}\}$.

3.

(i) (6 points) Construct a bijection $(0, 1) \to \mathbb{R}$.

(ii) (4 points) Construct an injection $(0, 1) \times (0, 1) \to (0, 1)$. Hint: elements of $(0, 1)$ can be represented as $0.a_1a_2\cdots$.

4.

(i) (8 points) Give pseudocode for an algorithm that finds all numbers appearing exactly twice in a sequence of length n, where all numbers are at most 1000.

(ii) (2 points) Directly state the number of comparisons used by your algorithm in the worst case, excluding comparisons in loop conditions.

5. (10 points) Show that if a and m are relatively prime positive integers, then the inverse of a modulo m is unique modulo m.

6. (10 points) Show that if a and b are positive integers, then

$$(2^a - 1) \bmod (2^b - 1) = 2^{a \bmod b} - 1.$$

7.

(i) (6 points) Compute $3^{2003} \pmod 7$.

(ii) (4 points) Compute the last digit in base 10 of $\lfloor 3^{2003}/7 \rfloor$.

8. (10 points) Use mathematical induction to show that

$$[(p_1 \to p_2) \wedge (p_2 \to p_3) \wedge \cdots \wedge (p_{n-1} \to p_n)] \to [(p_1 \wedge p_2 \wedge \cdots \wedge p_{n-1}) \to p_n]$$

is a tautology for all propositions $p_1, \ldots, p_n$, where $n \geq 2$.

9. (10 points) Which amounts of money can be formed using only two-dollar bills and five-dollar bills? Prove your answer using strong induction.

10.

(i) (8 points) Give iterative and recursive algorithms for finding the nth term of the sequence

$$a_0 = 1, \quad a_1 = 3, \quad a_2 = 5, \qquad a_n = a_{n-1} a_{n-2}^2 a_{n-3}^3 \quad (n \geq 3).$$

(ii) (2 points) Which algorithm is more efficient?

6 2025 Midterm Exam (LLF)

Discrete Mathematics, Spring & Summer 2025

Midterm Exam (100 pts)

1. (10 points) Determine whether the following statements are true or false.

( ) (1) If $a \equiv b \pmod{m}$ and $a \equiv c \pmod{m}$, then $a \equiv b + c \pmod{m}$.

( ) (2) An inverse of 2 modulo 31 is -15.

( ) (3) The number of ways to partition a set with 10 elements into two nonempty disjoint subsets is 512.

( ) (4) The following is a valid recursive definition of a function f from the set of nonnegative integers to the set of integers:

$$f(0) = 0, \qquad f(1) = 1, \qquad f(n) = 2f(n-1) \quad (n \geq 1).$$

( ) (5) Mathematical induction, strong induction, and well-ordering are three equivalent principles.

2. (30 points) Fill in the blanks.

(1) Use Fermat's little theorem to find $9^{45} \pmod{23}$. _____

(2) A professor gives a multiple-choice quiz with ten questions, each with four possible responses a, b, c, d. What is the minimum number of students needed to guarantee that at least three answer sheets are identical, assuming no answers are left blank? _____

(3) Suppose $|A| = 8$ and $|B| = 5$. Find the number of functions $f : A \to B$.

_____

How many of these functions are one-to-one? _____

(4) Find the number of permutations of the string `artificial-intelligence`.

_____

(5) How many integer solutions are there to $x_1 + x_2 + x_3 + x_4 = 20$, where $x_i \geq 2$ for $i = 1, 2, 3, 4$? _____

(6) Find the coefficient of x^{12} in the power series of $\dfrac{x^3}{(1+2x)^2}$. _____

(7) How many onto functions are there from a set with eight elements to a set with four elements? _____

(8) Find a recurrence relation for the number of ways to climb an n-step staircase if you may climb either one or two steps at a time. _____

(9) Suppose $|A| = n$. There are _____ distinct reflexive and symmetric relations on A.

3. (10 points) Let S be the subset of ordered pairs of integers defined recursively by

$$(0,0) \in S, \qquad \text{if } (a,b) \in S, \text{ then } (a+2,b+3) \in S \text{ and } (a+3,b+2) \in S.$$

(1) List the elements of S produced by the first three applications of the recursive definition.

(2) Use structural induction to show that $5 \mid a+b$ whenever $(a,b) \in S$.

4. (5 points) Express $\gcd(84,18)$ as a linear combination of 18 and 84.

5. (10 points) Use generating functions to solve the recurrence

$$a_n = 3a_{n-1} + 2^n + 5, \qquad a_0 = 1.$$

6. (15 points) Given the sequence

$$2, 5, 9, 14, 20, 27, 35, \ldots$$

(1) Find a recurrence relation for a_n and give the necessary initial condition(s).

(2) Find an explicit formula for a_n by solving the recurrence in part (1) via its characteristic equation.

(3) Write down a closed form for the generating function of $\{a_n\}$.